Development Team Project Report Team F Group 4.docx

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Airbnb listing prices in New York City: a regression and clustering analysis

Development Team Project – Project Report

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(Team F, Group 4)

Introduction

Airbnb provides short-term accommodation across many locations and price ranges in New York City. Location is an important factor in pricing (Deboosere et al., 2019). Therefore, we assume that distance from the most centrally located touristic area is associated with listing price. Times Square has been described as the "epicenter of NYC tourism" (Weaver, 2025). Thus, our research question is as follows:

> *How is distance from Times Square associated with Airbnb listing prices in New York City, and what listing segments can be identified based on location and property characteristics?*

We used regression and clustering to identify different types of Airbnb listings. The main purpose of this research is to generate insights that could help Airbnb understand pricing patterns and different listing segments in New York City.



Figure 1: Map of the five boroughs of New York City

Adapted from: https://ontheworldmap.com/usa/city/new-york-city/nyc-boroughs-map.html

Data Preparation

We cleaned the original 2019 Airbnb New York dataset (Dgomonov, 2019), containing 48,895 Airbnb listings. The cleaned version consisted of 48,884 listings. We added an extra column - distance_times_square_km, which calculates distance from Times Square using latitude and longitude.

Exploratory Data Analysis

We first visualised listing prices according to geographic location:

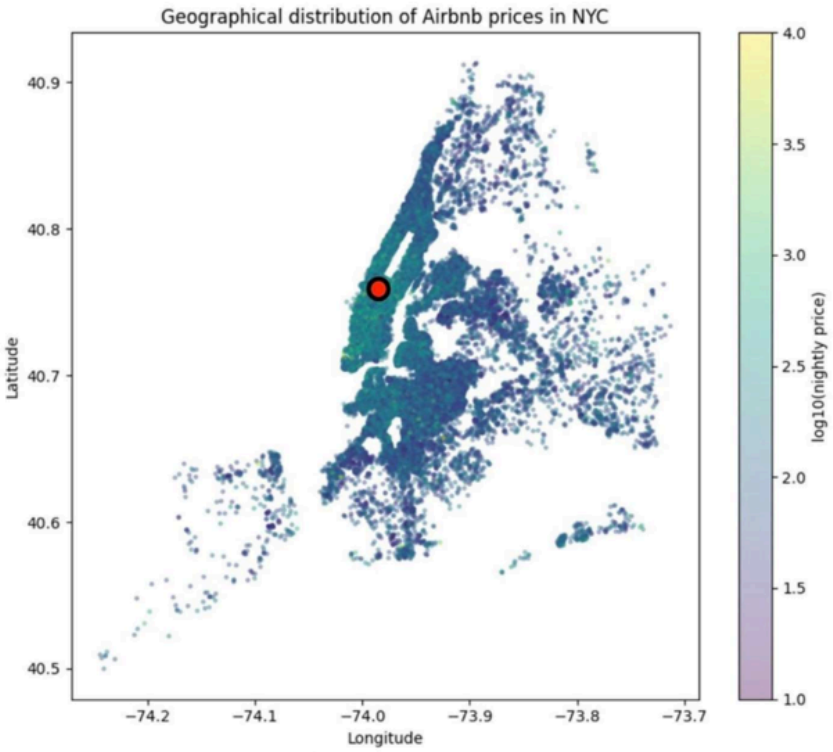


Figure 2: Geographical distribution of Airbnb prices in NYC

The red dot represents the location of Times Square. Higher prices are concentrated around Manhattan/Times Square. Listings further away are generally cheaper.

Next, we visualised the median Airbnb price by distance from Times Square:

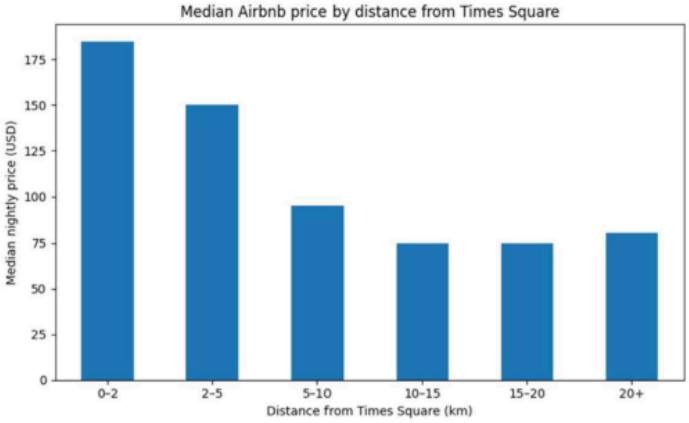


Figure 3: Median Airbnb price by distance from Times Square

Listings within 0-2 km of Times Square have a median price of $185. At 2-5 km, this falls to $150. At 5-10 km, it falls to $95.For a distance >10km, the price is around $75.

Evidently, Airbnb prices tend to decrease as distance from Times Square increases. Interestingly, prices slightly increase from 20+ km, making the relationship not perfectly straight.

However, distance from Times Square may not be the only factor influencing prices. Borough, room type, quality, neighbourhood and other characteristics may also have an influence. Therefore, regression and clustering are used to further investigate these relationships.

Regression

Through regression, we predicted the value of price based on how one or more variables change. We firstly decided to transform price into its logarithmic form log(price) to reduce the impact of outlying prices (5-10k $).

We checked multicollinearity amongst Distance and Borough. Numbers greater than 10 could represent severe multicollinearity, whilst lower numbers show reduced to no multicollinearity (Wooldridge, 2020). Distance had a Variance Inflation Factor (VIF) of 1.9. Borough analysis showed a VIF >10, which is high. Thus, the split of credit between Borough and Distance should not be taken as separate correlations.

Secondly, we split the data into a test and a training set, respectively containing 20% and 80% of the data. We used Borough to stratify the split so as to avoid underrepresenting the rarest labels and ensuring the data splits each contain proportionate amounts of the overall data (Géron, 2025)

We used three different regression approaches and compared them:

1. A simple linear regression on distance alone found each +1km was associated with a ~5.9% price drop (R^2=0.15).

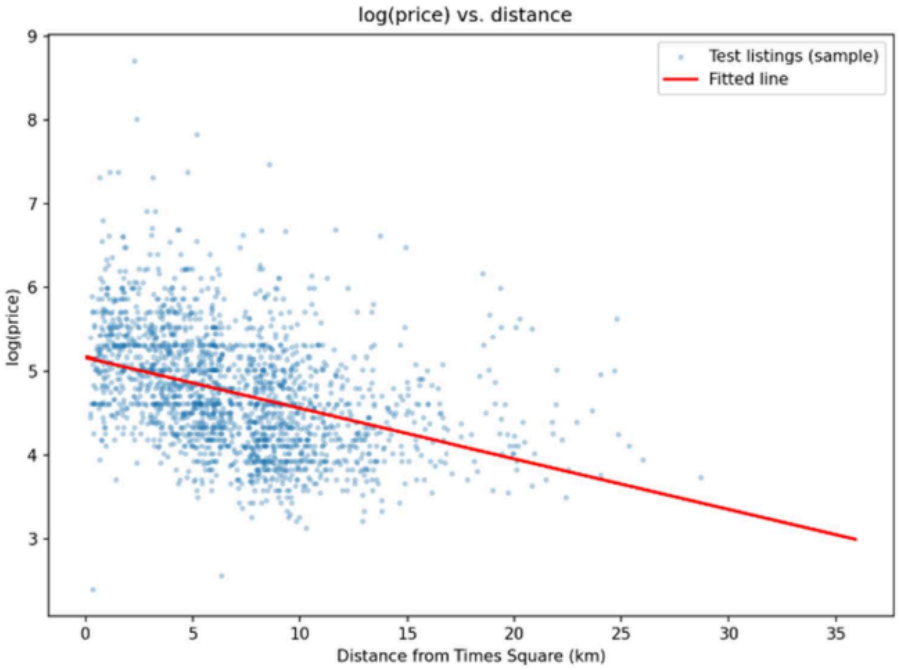


Figure 4: Log(price) vs. distance

2. A multilinear regression approach with all features; controlling for room type and borough caused distance's effect to shrink, showing it was probably mediated by borough and room type.

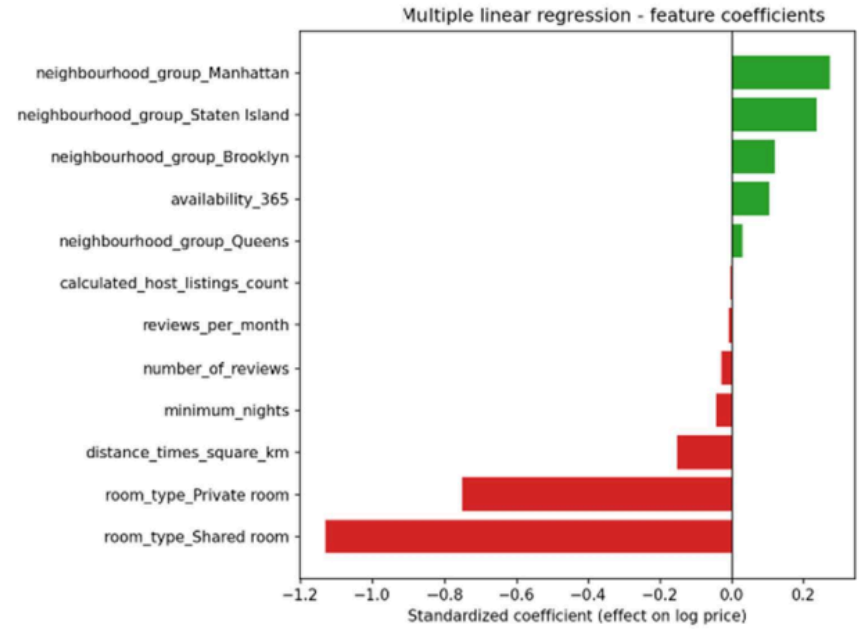


Figure 5: Multiple linear regression - feature coefficients

3. We also performed a polynomial regression approach to determine nonlinear relationships (Géron, 2025). We checked first with degree 2, which improved fit (R^2=0.51), revealing price drops steeply near the centre and then a flattening out. Degree 3 didn't add any significant improvement. In figure 6, degree 1 is the linear model.

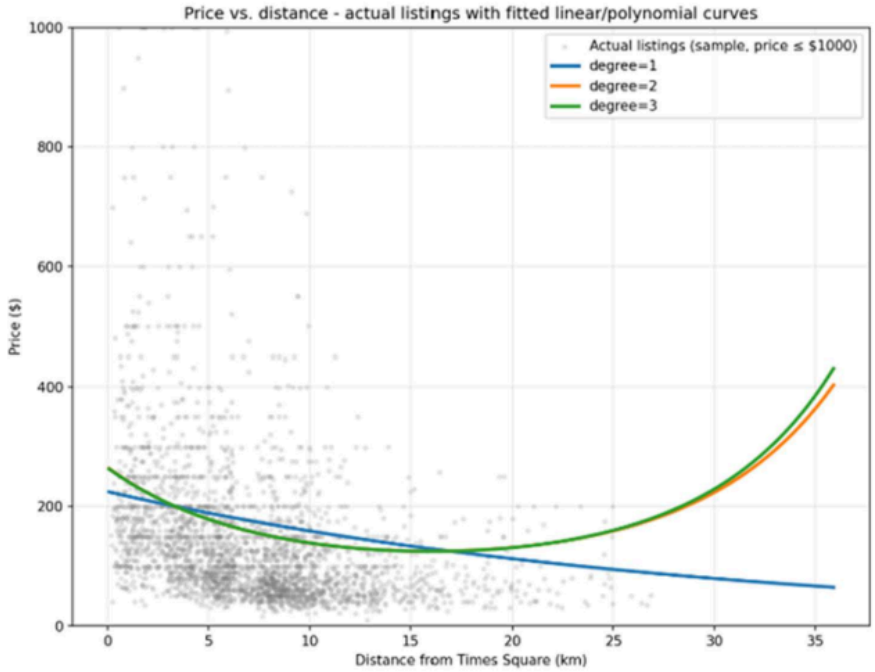


Figure 6: Price vs. distance – actual listings with fitted linear/polynomial curves

The degree-2 polynomial achieved a test RMSE of 0.488 (log-price units) and R^2 of 0.512, both an improvement over the plain linear model (RMSE 0.496, R^2 0.496). This confirms that the true

relationship between distance and price is not linear. Although the model predicts that price sky-rockets at some point, we feel this should be further investigated.

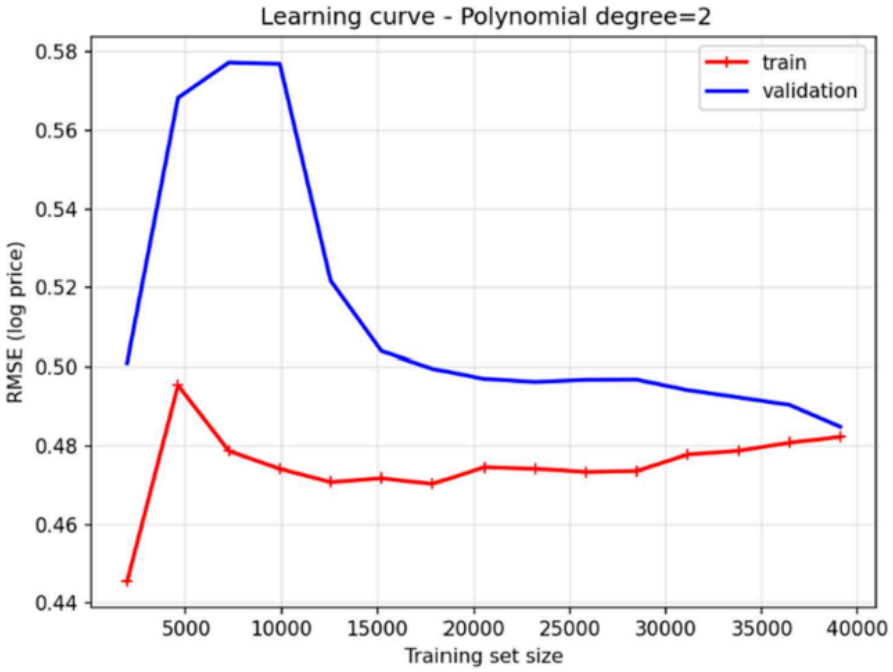


Figure 7: Learning curve – Polynomial degree = 2

Ultimately, the learning curves graph showed train/validation RMSE converging closely. This shows that the model fairly predicts price, without overfitting the data (Géron, 2025).

As there were no signs of overfitting, we chose not to use any regularisation technique on the model.

Clustering

Clustering was used to identify distinct groups of Airbnb listings with similar location and property characteristics. Géron (2025)

The optimal number of clusters was evaluated using the elbow method – identifying the point at which the data turns a corner and no longer grows. (Herdiana et al., 2025) - and average silhouette scores.

Although k = 2 produced the highest silhouette score (0.392 - from a range of -1 to +1), this solution was considered too coarse to provide meaningful segmentation. The three-cluster solution achieved the next highest silhouette score (0.252) and corresponded with the point at which improvements in within-cluster variation began to flatten; therefore, k = 3 was selected.

K-means clustering was performed on six scaled variables. Min-max scaling and standardisation techniques as described by Kelsy Cabello-Solorzano et al., (2023) were used. Figure 8 uses principal component analysis to visualise the resulting clusters in two dimensions. Clustering was not performed on the principal components themselves. Together, PC1 and PC2 explain 52.5% of the variance. The clusters occupy distinguishable regions of the PCA space, although some overlap remains.



Figure 8: PCA projection of K-means Airbnb listing clusters (k = 3)

Cluster profiling (Table 1) indicates that Cluster 1 represents a long-stay segment, with a 30-night median minimum stay, high availability (316 days) and hosts with a median of 16 listings . Cluster 2 represents short-stay listings, with the lowest median price ($95), greatest distance from Times Square (8.7 km) and highest review activity (2.12 reviews/month). Cluster 3 comprises the largest segment (54.6%), characterised by zero median availability and low review activity.

	Cluster 1	Cluster 2	Cluster 3
Size	4,594 (9.4%)	17,608 (36.0%)	26,682 (54.6%)
Median price	$150	$95	$115
Median distance from Times Sq.	3.9 km (closest)	8.7 km (furthest)	5.7 km (middle)
Median minimum nights	**30**	2	3
Median # reviews	1	28	2
Median reviews/month	0.09	2.12	0.12
Median availability (days/365)	**316**	197	**0**
Median host listings count	**16**	1	1
Dominant room type	Entire home/apt (68%)	Private room (54%)	Entire home/apt (55%)
Dominant borough	Manhattan (70%)	Brooklyn (44%), spread across boroughs	Manhattan (49%) / Brooklyn (43%)

Table 1: Profile Characteristics of the Three K-Means Airbnb Listing Clusters

Similarly to regression, cluster prices decline with increasing distance from Times Square ($150, $115 and $95), but listing segments also differ in host, availability and booking characteristics. However, the moderate silhouette score and overlap in the PCA projection indicate that these segments should be interpreted as broad listing profiles rather than sharply separated groups.

Conclusion

Listings further from Times Square tend to be cheaper, although the relationship isn't perfectly linear. But borough, room type and other listing characteristics also explain a large part of price differences. The clustering backed this up, revealing three broad types of listings with different prices, availability, host behaviour and review activity. Overall, Airbnb pricing reflects a mix of location and listing characteristics, rather than any single factor.

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